Linear System — Homework 4

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Exercise 1

Determine whether or not the following matrix is nondefective.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

Solution

$$\begin{aligned} \det(A - sI) &= \begin{vmatrix} -s & 1 & 0 \\ 0 & -s & 1 \\ -1 & 1 & 1-s \end{vmatrix} \\ &= -s\Big(s(s-1) - 1\Big) + (-1) \cdot 1 \\ &= -s(s^2 - s - 1) - 1 \\ &= -s^3 + s^2 + s - 1 \\ &= -(s-1)^2(s+1) \end{aligned}$$

for $s = 1$

$$A - sI = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Hence, for eigenvalue $s = 1$, AM $= 2$, and GM $= 1$

for $s = -1$

$$A - sI = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \Rightarrow \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Hence, for eigenvalue $s = -1$, AM $= 1$, and GM $= 1$

Therefore, matrix A is defective.

Exercise 2

What is the companion matrix whose eigenvalues are -1, -2, and -3?

Solution

$$\begin{aligned} p(s) &= (s+1)(s+2)(s+3) \\ &= a_3 s^3 + a_2 s^2 + a_1 s + a_0 \\ &= s^3 + 6s^2 + 11s + 6 \end{aligned}$$

Hence, companion matrix is

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \end{aligned}$$

Exercise 3

What is the real 2×2 companion matrix with eigenvalues $1 + j$, $1 - j$?

Solution

$$\begin{aligned} p(s) &= (s - (1+j))(s - (1-j)) \\ &= a_2 s^2 + a_1 s + a_0 \\ &= s^2 - 2s + 2 \end{aligned}$$

Hence, companion matrix is

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -2 & 2 \end{bmatrix} \end{aligned}$$

Exercise 4

Suppose A is a real square matrix and the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix}, \qquad \begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -1, 2, and j, respectively. What is the response $x(t)$ of the system

$$\dot{x} = Ax$$

to the following initial conditions?

(a) $x(0) = \begin{pmatrix} 2 \\ -2 \\ 2 \\ -2 \end{pmatrix}$

Solution

Since $\mathbf{v}_1$ is an eigenvector corresponding to eigenvalue $\lambda_1 = -1$,

$$x(0) = \begin{bmatrix} 2 \\ -2 \\ 2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = 2\mathbf{v}_1 \triangleq \mathbf{v}$$

$$A\mathbf{v} = \lambda_1 \mathbf{v}, \quad x(0) = \mathbf{v} \quad \Rightarrow \quad x(t) = e^{\lambda_1 t}\mathbf{v}$$

$$\begin{aligned} x(t) &= e^{\lambda_1 t} x(0) \\ &= e^{\lambda_1 t}\mathbf{v} \\ &= e^{-t} \cdot 2\mathbf{v}_1 \\ &= \begin{bmatrix} 2e^{-t} \\ -2e^{-t} \\ 2e^{-t} \\ -2e^{-t} \end{bmatrix}. \end{aligned}$$

(b) $x(0) = \begin{pmatrix} -1 \\ -2 \\ -4 \\ -8 \end{pmatrix}$

Solution

$$x(0) = \begin{bmatrix} -1 \\ -2 \\ -4 \\ -8 \end{bmatrix} = -\begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} = -\mathbf{v}_2 \text{ which corresponding to eigenvalue is } 2$$

$$\mathbf{v} := x(0) = -\mathbf{v}_2, \quad A\mathbf{v} = 2\mathbf{v} \quad \Rightarrow \quad x(t) = e^{2t}\mathbf{v}$$

$$\begin{aligned} x(t) &= e^{2t}x(0) \\ &= e^{2t}\mathbf{v} \\ &= e^{2t}\cdot -\mathbf{v}_2 \\ &= e^{2t}\begin{bmatrix}-1\\-2\\-4\\-8\end{bmatrix} = \begin{bmatrix}-e^{2t}\\-2e^{2t}\\-4e^{2t}\\-8e^{2t}\end{bmatrix} \end{aligned}$$

(c) $x(0) = \begin{pmatrix}0\\3\\3\\9\end{pmatrix}$

Solution

$$x(0) = \mathbf{v} = \begin{bmatrix}0\\3\\3\\9\end{bmatrix} = -\begin{bmatrix}1\\-1\\1\\-1\end{bmatrix} + \begin{bmatrix}1\\2\\4\\8\end{bmatrix} = -\mathbf{v}_1 + \mathbf{v}_2$$

with linear combination

$$\begin{aligned} x(t) &= -e^{\lambda_1 t}\mathbf{v}_1 + e^{\lambda_2 t}\mathbf{v}_2 \\ &= \begin{bmatrix}-e^{-t}+e^{2t}\\e^{-t}+2e^{2t}\\-e^{-t}+4e^{2t}\\e^{-t}+8e^{2t}\end{bmatrix} \end{aligned}$$

(d) $x(0) = \begin{pmatrix}1\\1\\-1\\-1\end{pmatrix}$

Solution

$$\mathbf{v}_3 = \mathbf{u}_3 + j\mathbf{p}_3, \qquad \mathbf{u}_3 = \begin{bmatrix}1\\0\\-1\\0\end{bmatrix}, \qquad \mathbf{p}_3 = \begin{bmatrix}0\\1\\0\\-1\end{bmatrix}.$$

$$\lambda_3 = \alpha + j\omega = j \Rightarrow \alpha = 0, \quad \omega = 1$$

$$
\begin{aligned}
x(0) &= \mathbf{u} + \mathbf{p} \\
x(t) &= x_u(t) + x_p(t) \\
&= e^{\alpha t}\Big(\cos(\omega t) \cdot \mathbf{u} - \sin(\omega t) \cdot \mathbf{p} \Big) \\
&\quad + e^{\alpha t}\Big(\sin(\omega t) \cdot \mathbf{u} + \cos(\omega t) \cdot \mathbf{p} \Big)
\end{aligned}
$$

Here $\mathbf{u} = \mathbf{u}_3$, $\mathbf{p} = \mathbf{p}_3$, $\alpha = 0$, and $\omega = 1$.

$$
\begin{aligned}
x(t) &= \mathbf{u} \cdot (\cos t + \sin t) + \mathbf{p} \cdot (\cos t - \sin t) \\
&= \begin{bmatrix} \cos t + \sin t \\ \cos t - \sin t \\ -\cos t - \sin t \\ -\cos t + \sin t \end{bmatrix}
\end{aligned}
$$

Exercise 5

Consider a discrete-time LTI system described by

$$x(k+1) = Ax(k).$$

(a) Suppose that the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

are eigenvectors of A corresponding to eigenvalues -2 and 3, respectively. What is the response $x(k)$ of the system to the initial condition

$$x(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}?$$

Solution

$$x(0) = \frac{1}{2}\left(\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right) = \mathbf{v} = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$$

$$
\begin{aligned}
x(k) &= A^k \left(\frac{1}{2}\mathbf{v}_1 + \frac{1}{2}\mathbf{v}_2 \right) \\
&= \frac{1}{2}A^k\mathbf{v}_1 + \frac{1}{2}A^k\mathbf{v}_2 \\
&= \frac{1}{2}\lambda_1^k\mathbf{v}_1 + \frac{1}{2}\lambda_2^k\mathbf{v}_2 \\
&= \frac{1}{2}(-2)^k \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + \frac{1}{2}(3)^k \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \\
&= \frac{1}{2} \begin{bmatrix} (-2)^k + 3^k \\ -(-2)^k + 3^k \\ (-2)^k + 3^k \\ -(-2)^k + 3^k \end{bmatrix}
\end{aligned}
$$

(b) Suppose A is a real matrix and the vector

$$
\begin{pmatrix} 1 \\ j \\ -1 \\ -j \end{pmatrix}
$$

is an eigenvector of A corresponding to the eigenvalue $2 + 3j$. What is the response $x(k)$ of the system to the initial condition

$$
x(0) = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}?
$$

Solution

$$
\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} + j \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \mathbf{u} + j\mathbf{p}
$$

$$
x(0) = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} = \mathrm{Re}(\mathbf{v}) = \mathbf{u}
$$

Let $r = \sqrt{13}$ and $\theta = \arctan(3/2)$, so that $2 + 3j = r(\cos\theta + j\sin\theta)$. Since A is real,

$$
\begin{aligned}
x(k) &= A^k x(0) = A^k \,\mathrm{Re}(\mathbf{v}) \\
&= \mathrm{Re}(A^k\mathbf{v}) = \mathrm{Re}\big((2+3j)^k\mathbf{v}\big) \\
&= r^k \,\mathrm{Re}\big[(\cos(k\theta) + j\sin(k\theta))\mathbf{v}\big] \\
&= r^k \,\mathrm{Re}\big[(\cos(k\theta) + j\sin(k\theta))(\mathbf{u} + j\mathbf{p})\big] \\
&= r^k \big(\cos(k\theta)\mathbf{u} - \sin(k\theta)\mathbf{p}\big).
\end{aligned}
$$

$$x(k) = \sqrt{13}^k \left(\cos(k \arctan(3/2)) \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} - \sin(k \arctan(3/2)) \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right)$$
$$= \begin{bmatrix} \sqrt{13}^k \cos(k \arctan(3/2)) \\ -\sqrt{13}^k \sin(k \arctan(3/2)) \\ -\sqrt{13}^k \cos(k \arctan(3/2)) \\ \sqrt{13}^k \sin(k \arctan(3/2)) \end{bmatrix}$$

Exercise 6

Consider a 5×5 matrix

$$A = vw^T$$

where v and w are 5×1 matrices.

(a) What is the rank of A?

Solution

for $v, w \neq 0$

$$A = vw^T = \begin{bmatrix} w_1 v & w_2 v & w_3 v & w_4 v & w_5 v \end{bmatrix}$$

Hence,

$$\text{range}(A) = \text{span}\{v\}$$

thus,

$$\text{rank}\, A = \dim(\text{range}(A)) = 1$$

but if v or w is 0, $\text{rank}\, A = 0$

(b) What is the nullity of A?

Solution

If $v \neq 0$ and $w \neq 0$,

$$\text{nullity}(A) = 5 - \text{rank}(A) = 5 - 1 = 4$$

else

$$\text{nullity}(A) = 5$$

(c) What are the eigenvalues of A if $w^T v \neq 0$?

Solution

$$A = vw^T \Rightarrow Av = v(w^T v)_{1\times 1} = (w^T v)v = \lambda v$$

Hence, the eigenvalue is $w^T v$

Since the nullity of A is 4, there are four linearly independent eigenvectors corresponding to eigenvalue 0. Therefore, the eigenvalues are

$$w^T v, \quad 0, \quad 0, \quad 0, \quad 0$$

For Exercise 7 and 8, please use Matlab/Python to illustrate the time history plot. Please don't plot y, θ_1, θ_2 in the same plot, separate them!

Exercise 7

Recall the 2 pendulum cart system.

$$\begin{aligned}
&(m_0 + m_1 + m_2)\ddot{y} - m_1 l_1 \cos\theta_1\, \ddot{\theta}_1 - m_2 l_2 \cos\theta_2\, \ddot{\theta}_2 \\
&\qquad + m_1 l_1 \sin\theta_1\, \dot{\theta}_1^2 + m_2 l_2 \sin\theta_2\, \dot{\theta}_2^2 = u \\
&- m_1 l_1 \cos\theta_1\, \ddot{y} + m_1 l_1^2 \ddot{\theta}_1 + m_1 l_1 g \sin\theta_1 = 0 \\
&- m_2 l_2 \cos\theta_2\, \ddot{y} + m_2 l_2^2 \ddot{\theta}_2 + m_2 l_2 g \sin\theta_2 = 0
\end{aligned}$$

Consider the equilibrium configurations defined by $u^e = 0$ and

$$E1: \qquad (y^e, \theta_1^e, \theta_2^e) = (0, 0, 0)$$

and

$$E2: \qquad (y^e, \theta_1^e, \theta_2^e) = (0, \pi, \pi).$$

Consider state space representations of the linearizations corresponding to the following combinations of parameters and equilibrium conditions:

	Parameter	Equilibrium
$L1$	$P1$	$E1$
$L2$	$P1$	$E2$
$L3$	$P2$	$E1$
$L4$	$P2$	$E2$
$L5$	$P3$	$E1$
$L6$	$P3$	$E2$
$L7$	$P4$	$E1$
$L8$	$P4$	$E2$

(a) Determine the eigenvalues of all the ***linearized systems*** L1–L8.

Solution

E1: $(y^e, \theta_1^e, \theta_2^e) = (0, 0, 0)$

Construction of the state-space model. Near the equilibrium

$$\sin\theta_i \approx \theta_i, \quad \cos\theta_i \approx 1$$

The terms $\sin\theta_i\,\dot{\theta}_i^2$ are neglected because they are higher-order terms in the perturbations. The linearized equations become

$$\begin{cases} (m_0+m_1+m_2)\ddot{y} - m_1 l_1 \ddot{\theta}_1 - m_2 l_2 \ddot{\theta}_2 & = u \\ -m_1 l_1 \ddot{y} + m_1 l_1^2 \ddot{\theta}_1 + m_1 l_1 g \theta_1 & = 0 \\ -m_2 l_2 \ddot{y} + m_2 l_2^2 \ddot{\theta}_2 + m_2 l_2 g \theta_2 & = 0 \end{cases}$$

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} y \\ \theta_1 \\ \theta_2 \\ \dot{y} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, \Rightarrow \dot{x} = \begin{bmatrix} \dot{y} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{y} \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} x_4 \\ x_5 \\ x_6 \\ \ddot{y} \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}$$

obtained with rewritten equation

$$\ddot{\theta}_1 = \frac{\ddot{y} - g\theta_1}{l_1}, \text{ and } \ddot{\theta}_2 = \frac{\ddot{y} - g\theta_2}{l_2}$$

Substituting into the first equation gives

$$\begin{aligned} u &= (m_0+m_1+m_2)\ddot{y} - m_1(\ddot{y} - g\theta_1) - m_2(\ddot{y} - g\theta_2) \\ &= m_0\ddot{y} + m_1 g\theta_1 + m_2 g\theta_2. \end{aligned}$$

$$\ddot{y} = \frac{-m_1 g\theta_1 - m_2 g\theta_2 + u}{m_0}$$

Hence,

$$\begin{aligned} \ddot{\theta}_1 &= \frac{\ddot{y} - g\theta_1}{l_1} = \frac{-(m_0+m_1)g\theta_1 - m_2 g\theta_2 + u}{m_0 l_1}, \\ \ddot{\theta}_2 &= \frac{\ddot{y} - g\theta_2}{l_2} = \frac{-m_1 g\theta_1 - (m_0+m_2)g\theta_2 + u}{m_0 l_2}. \end{aligned}$$

$$
\begin{aligned}
\dot{x} = \begin{bmatrix} x_4 \\ x_5 \\ x_6 \\ \ddot{y} \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} &= \begin{bmatrix} x_4 \\ x_5 \\ x_6 \\ \frac{-m_1 g}{m_0} x_2 - \frac{m_2 g}{m_0} x_3 + \frac{1}{m_0} u \\ \frac{-(m_0+m_1)g}{m_0 l_1} x_2 - \frac{m_2 g}{m_0 l_1} x_3 + \frac{1}{m_0 l_1} u \\ \frac{-m_1 g}{m_0 l_2} x_2 - \frac{(m_0+m_2)g}{m_0 l_2} x_3 + \frac{1}{m_0 l_2} u \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{-m_1 g}{m_0} & \frac{-m_2 g}{m_0} & 0 & 0 & 0 \\ 0 & \frac{-(m_0+m_1)g}{m_0 l_1} & \frac{-m_2 g}{m_0 l_1} & 0 & 0 & 0 \\ 0 & \frac{-m_1 g}{m_0 l_2} & \frac{-(m_0+m_2)g}{m_0 l_2} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_0} \\ \frac{1}{m_0 l_1} \\ \frac{1}{m_0 l_2} \end{bmatrix} u \\
&= Ax + Bu
\end{aligned}
$$

L1 (P1) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 1, 1)$

$$
\begin{aligned}
A &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{-m_1 g}{m_0} & \frac{-m_2 g}{m_0} & 0 & 0 & 0 \\ 0 & \frac{-(m_0+m_1)g}{m_0 l_1} & \frac{-m_2 g}{m_0 l_1} & 0 & 0 & 0 \\ 0 & \frac{-m_1 g}{m_0 l_2} & \frac{-(m_0+m_2)g}{m_0 l_2} & 0 & 0 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{-1}{2} & \frac{-1}{2} & 0 & 0 & 0 \\ 0 & \frac{-3}{2} & \frac{-1}{2} & 0 & 0 & 0 \\ 0 & \frac{-1}{2} & \frac{-3}{2} & 0 & 0 & 0 \end{bmatrix}
\end{aligned}
$$

```
A = [
0    0     0    1    0    0;
0    0     0    0    1    0;
0    0     0    0    0    1;
0  -1/2  -1/2   0    0    0;
0  -3/2  -1/2   0    0    0;
0  -1/2  -3/2   0    0    0
];

lambda = eig(A)
```

MATLAB output:

```
lambda =

   0.0000 + 0.0000i
   0.0000 + 0.0000i
   0.0000 + 1.4142i
   0.0000 - 1.4142i
  -0.0000 + 1.0000i
  -0.0000 - 1.0000i
```

The displayed -0.0000 values are numerical roundoff.

Hence, the eigenvalues (including multiplicities) are $0, 0, \pm j, \pm\sqrt{2}j$.

L3 (P2) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.99, 1)$

$$A_{L3} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & -\frac{3}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & -\frac{50}{99} & -\frac{50}{33} & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/2 0 0 0;
0 -3/2 -1/2 0 0 0;
0 -50/99 -50/33 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

   0.0000 + 0.0000i
   0.0000 + 0.0000i
  -0.0000 + 1.4178i
  -0.0000 - 1.4178i
   0.0000 + 1.0025i
   0.0000 - 1.0025i
```

Hence, the eigenvalues are approximately $0, 0, \pm 1.0025j, \pm 1.4178j$.

L5 (P3) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 0.5, 1, 1, 1)$

$$A_{L5} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{4} & 0 & 0 & 0 \\ 0 & -\frac{3}{2} & -\frac{1}{4} & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{5}{4} & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/4 0 0 0;
0 -3/2 -1/4 0 0 0;
0 -1/2 -5/4 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

   0.0000 + 0.0000i
   0.0000 + 0.0000i
  -0.0000 + 1.3229i
  -0.0000 - 1.3229i
   0.0000 + 1.0000i
   0.0000 - 1.0000i
```

Hence, the eigenvalues are approximately $0, 0, \pm 1.0000j, \pm 1.3229j$.

L7 (P4) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.5, 1)$

$$A_{L7} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & -\frac{3}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & -1 & -3 & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/2 0 0 0;
0 -3/2 -1/2 0 0 0;
0 -1 -3 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

   0.0000 + 0.0000i
   0.0000 + 0.0000i
  -0.0000 + 1.8113i
  -0.0000 - 1.8113i
   0.0000 + 1.1042i
   0.0000 - 1.1042i
```

Hence, the eigenvalues are approximately $0, 0, \pm 1.1042j, \pm 1.8113j$.

E2: $(y^e, \theta_1^e, \theta_2^e) = (0, \pi, \pi)$

Construction of the state-space model. Let $\delta\theta_i = \theta_i - \pi$ and $\delta u = u - u^e = u$. Near E2,

$$\sin\theta_i \approx -\delta\theta_i, \qquad \cos\theta_i \approx -1.$$

Neglecting higher-order terms in the perturbations gives

$$\begin{cases} (m_0 + m_1 + m_2)\delta\ddot{y} + m_1 l_1 \delta\ddot{\theta}_1 + m_2 l_2 \delta\ddot{\theta}_2 = \delta u, \\ m_1 l_1 \delta\ddot{y} + m_1 l_1^2 \delta\ddot{\theta}_1 - m_1 l_1 g \delta\theta_1 = 0, \\ m_2 l_2 \delta\ddot{y} + m_2 l_2^2 \delta\ddot{\theta}_2 - m_2 l_2 g \delta\theta_2 = 0. \end{cases}$$

As in E1, solve the last two equations for the angular accelerations and substitute into the first equation:

$$\delta\ddot{\theta}_i = \frac{-\delta\ddot{y} + g\delta\theta_i}{l_i}, \qquad i = 1, 2,$$
$$\delta\ddot{y} = \frac{-m_1 g\delta\theta_1 - m_2 g\delta\theta_2 + \delta u}{m_0}.$$

Thus,

$$\delta\ddot{\theta}_1 = \frac{(m_0 + m_1)g\delta\theta_1 + m_2 g\delta\theta_2 - \delta u}{m_0 l_1},$$
$$\delta\ddot{\theta}_2 = \frac{m_1 g\delta\theta_1 + (m_0 + m_2)g\delta\theta_2 - \delta u}{m_0 l_2}.$$

Define the perturbation state

$$\delta x = x - x^e = \begin{bmatrix} y & \theta_1 - \pi & \theta_2 - \pi & \dot{y} & \dot{\theta}_1 & \dot{\theta}_2 \end{bmatrix}^T.$$

Then $\delta\dot{x} = A_{E2}\delta x + B_{E2}\delta u$, where

$$A_{E2} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{m_1 g}{m_0} & -\frac{m_2 g}{m_0} & 0 & 0 & 0 \\ 0 & \frac{(m_0+m_1)g}{m_0 l_1} & \frac{m_2 g}{m_0 l_1} & 0 & 0 & 0 \\ 0 & \frac{m_1 g}{m_0 l_2} & \frac{(m_0+m_2)g}{m_0 l_2} & 0 & 0 & 0 \end{bmatrix}, \qquad B_{E2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_0} \\ -\frac{1}{m_0 l_1} \\ -\frac{1}{m_0 l_2} \end{bmatrix}.$$

L2 (P1) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 1, 1)$

$$A_{L2} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/2 0 0 0;
0 3/2 1/2 0 0 0;
0 1/2 3/2 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

         0
         0
   -1.4142
   -1.0000
    1.4142
    1.0000
```

Hence, the eigenvalues are $0, 0, \pm 1, \pm\sqrt{2}$.

L4 (P2) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.99, 1)$

$$A_{L4} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{50}{99} & \frac{50}{33} & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/2 0 0 0;
0 3/2 1/2 0 0 0;
0 50/99 50/33 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

         0
         0
    1.4178
    1.0025
   -1.4178
   -1.0025
```

Hence, the eigenvalues are approximately $0, 0, \pm 1.0025, \pm 1.4178$.

L6 (P3) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 0.5, 1, 1, 1)$

$$A_{L6} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{5}{4} & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/4 0 0 0;
0 3/2 1/4 0 0 0;
0 1/2 5/4 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

         0
         0
   -1.3229
   -1.0000
    1.0000
    1.3229
```

Hence, the eigenvalues are $0, 0, \pm 1, \pm\sqrt{7}/2$.

L8 (P4) — $(m_0, m_1, m_2, l_1, l_2, g) = (2, 1, 1, 1, 0.5, 1)$

$$A_{L8} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 & 0 \end{bmatrix}.$$

```
A = [
0 0 0 1 0 0;
0 0 0 0 1 0;
0 0 0 0 0 1;
0 -1/2 -1/2 0 0 0;
0 3/2 1/2 0 0 0;
0 1 3 0 0 0
];
format short
lambda = eig(A)
```

MATLAB output:

```
lambda =

         0
         0
    1.8113
    1.1042
   -1.8113
   -1.1042
```

Hence, the eigenvalues are approximately $0, 0, \pm 1.1042, \pm 1.8113$.

(b) Compare the behavior of the nonlinear system with that of the linearized system for cases L7 and L8. Illustrate your results using different initial condition with time histories of y, θ_1, and θ_2. Plesae write paragraph to analysis the comparison between nonlinear and linearized system.

Initial conditions for case L7: ± 0 deg, ± 1 deg, ± 10 deg, ± 90 deg, ± 180 deg.

Initial conditions for case L8: ± 1 deg.

Solution

Comparison. For L7, the linearized and nonlinear responses agree closely for small angular deviations, while larger deviations produce greater differences because higher-order terms become important. At 180°, the nonlinear system remains at another equilibrium, which the model linearized about E1 does not capture. For L8, even a small initial deviation grows due to instability, so the linear approximation loses accuracy as the motion moves away from E2.

MATLAB time histories

Parameters P1–P4 follow Homework 1. The two pendulum lengths are l_1 and l_2; for P4, $l_1 = 1$, $l_2 = 0.5$.

State order: $x = [y, \theta_1, \theta_2, \dot{y}, \dot{\theta}_1, \dot{\theta}_2]^T$. Both models start from the same physical state $x(0) = x^e + z(0)$, with $u = 0$. Plots show deviations $z = x - x^e$; angles are in degrees and are not wrapped.

For Exercise 7(b), $z(0) = [0, -\alpha, +\alpha, 0, 0, 0]^T$ (angular entries converted from degrees to radians). L7 uses $\alpha = 0, 1, 10, 90, 180$ deg and L8 uses $\alpha = 1$ deg. L7 is simulated for 30 s and L8 for 5 s.

Nonlinear simulation: `ode45`, relative tolerance 10^{-10}, absolute tolerance 10^{-12}, maximum step 0.02 s. Linear responses use the matrix exponential.

L7: 0°

Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0, -0, 0, 0, 0, 0]
```

$x^e = [0, 0, 0, 0, 0, 0]^T$, $x(0) = x^e + z(0)$.

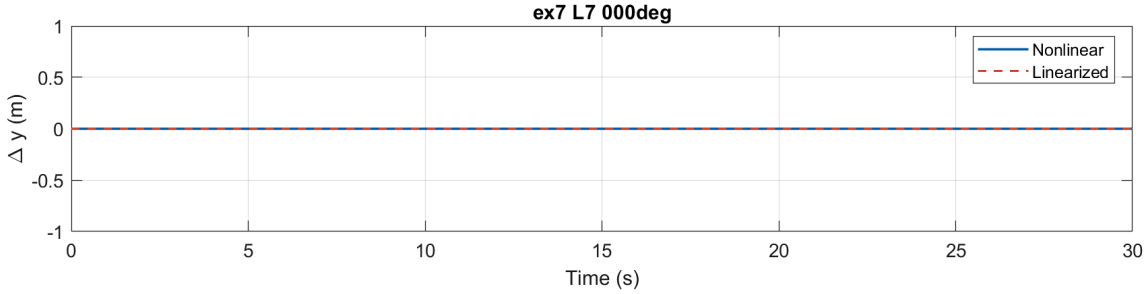


Figure 1: Δy: nonlinear and linearized time histories.

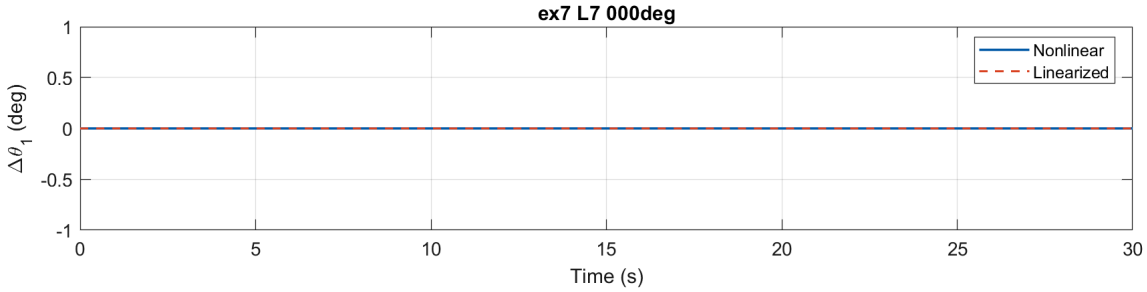


Figure 2: $\Delta\theta_1$: nonlinear and linearized time histories.

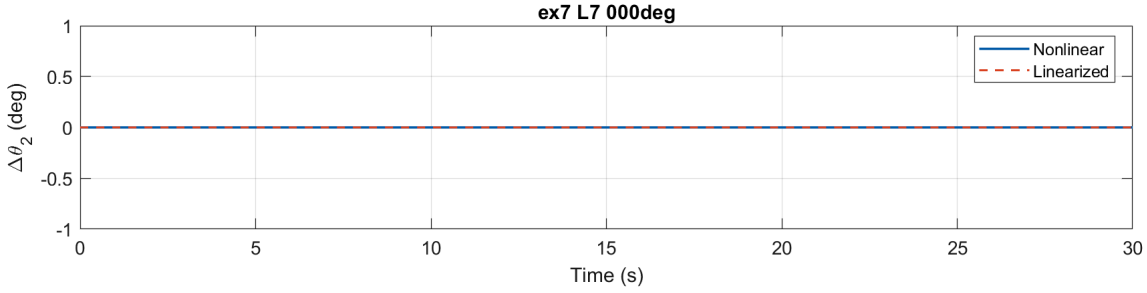


Figure 3: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code for L7, 0°: The following case-specific block uses the common local functions `simulate_case` and `export_case` listed in the source appendix.

```
p = [2 1 1 1 0.5 1]; % [m0 m1 m2 l1 l2 g]
model = cart_pendulum_model(p, 1);
alpha = 0; % degrees
z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];
options = odeset('RelTol',1e-10,'AbsTol',1e-12,'MaxStep',0.02);
r = simulate_case(model, z0, 30, 'ex7_L7_000deg', options);
export_case(r, outputDir, rootDir);
```

L7: $\pm 1^\circ$

Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0, -0.01745329, 0.01745329, 0, 0, 0]
```

$x^e = [0, 0, 0, 0, 0, 0]^T$, $x(0) = x^e + z(0)$.

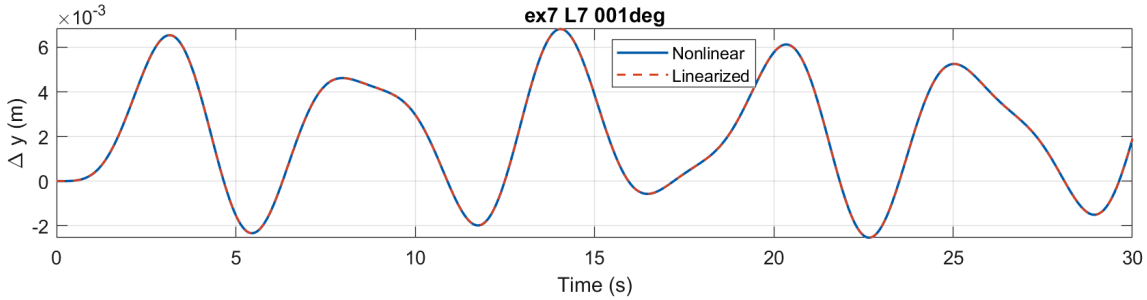


Figure 4: Δy: nonlinear and linearized time histories.

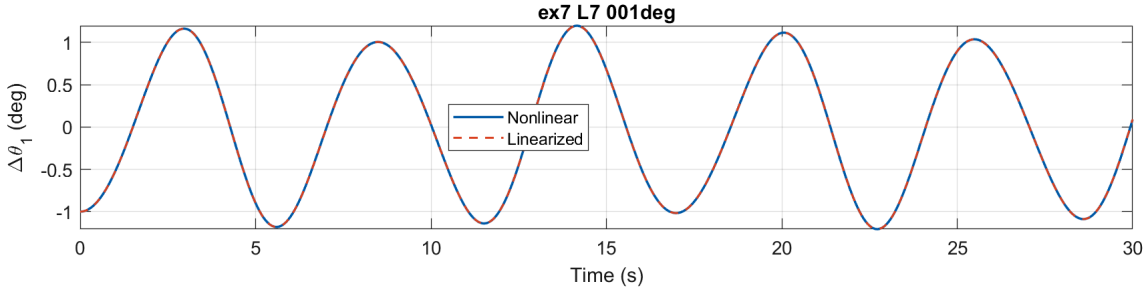


Figure 5: $\Delta\theta_1$: nonlinear and linearized time histories.

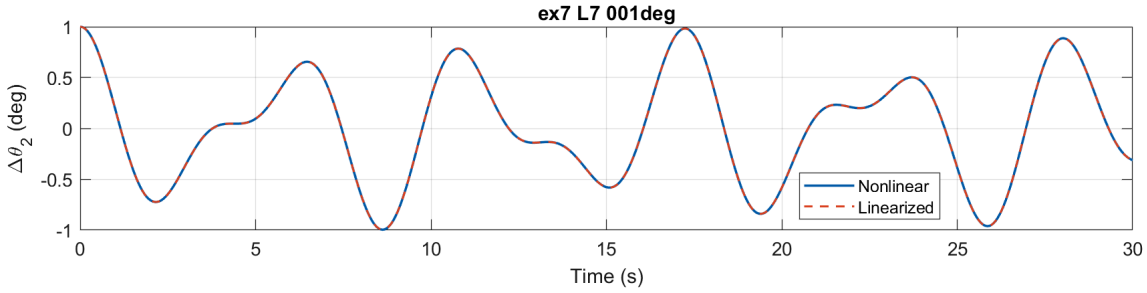


Figure 6: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code for L7, $\pm 1^\circ$: The following case-specific block uses the common local functions `simulate_case` and `export_case` listed in the source appendix.

```
p = [2 1 1 1 0.5 1]; % [m0 m1 m2 l1 l2 g]
model = cart_pendulum_model(p, 1);
alpha = 1; % degrees
z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];
options = odeset('RelTol',1e-10,'AbsTol',1e-12,'MaxStep',0.02);
r = simulate_case(model, z0, 30, 'ex7_L7_001deg', options);
export_case(r, outputDir, rootDir);
```

L7: $\pm 10^\circ$

Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0, -0.1745329, 0.1745329, 0, 0, 0]
```

$x^e = [0, 0, 0, 0, 0, 0]^T$, $x(0) = x^e + z(0)$.

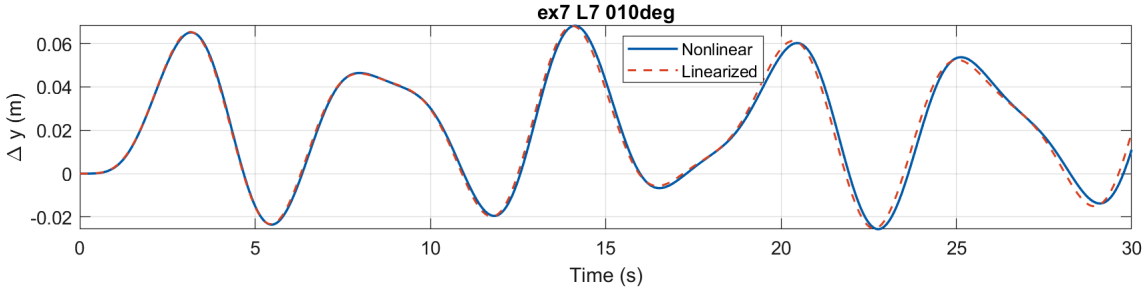


Figure 7: Δy: nonlinear and linearized time histories.

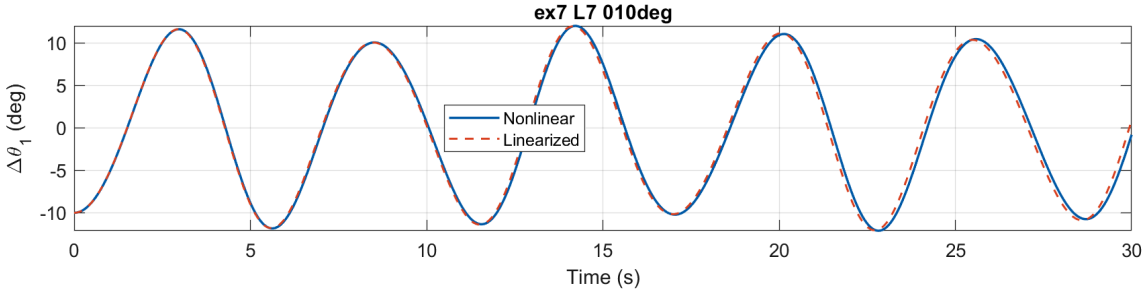


Figure 8: $\Delta\theta_1$: nonlinear and linearized time histories.

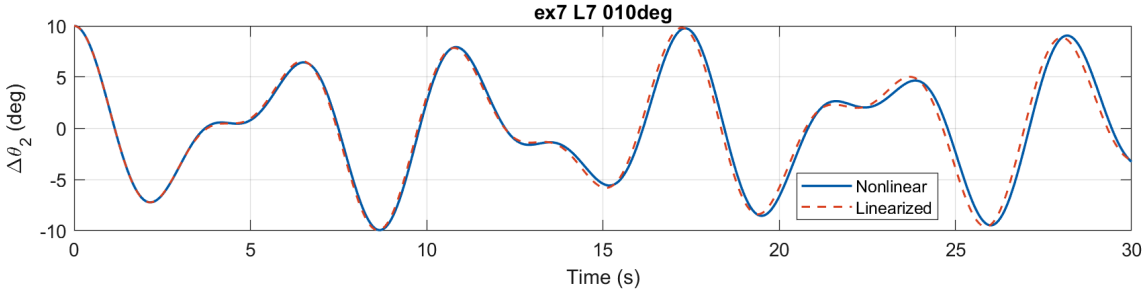


Figure 9: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code for L7, $\pm 10^\circ$: The following case-specific block uses the common local functions `simulate_case` and `export_case` listed in the source appendix.

```
p = [2 1 1 1 0.5 1]; % [m0 m1 m2 l1 l2 g]
model = cart_pendulum_model(p, 1);
alpha = 10; % degrees
z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];
options = odeset('RelTol',1e-10,'AbsTol',1e-12,'MaxStep',0.02);
r = simulate_case(model, z0, 30, 'ex7_L7_010deg', options);
export_case(r, outputDir, rootDir);
```

L7: $\pm 90^\circ$

Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0, -1.570796, 1.570796, 0, 0, 0]
```

$x^e = [0, 0, 0, 0, 0, 0]^T$, $x(0) = x^e + z(0)$.

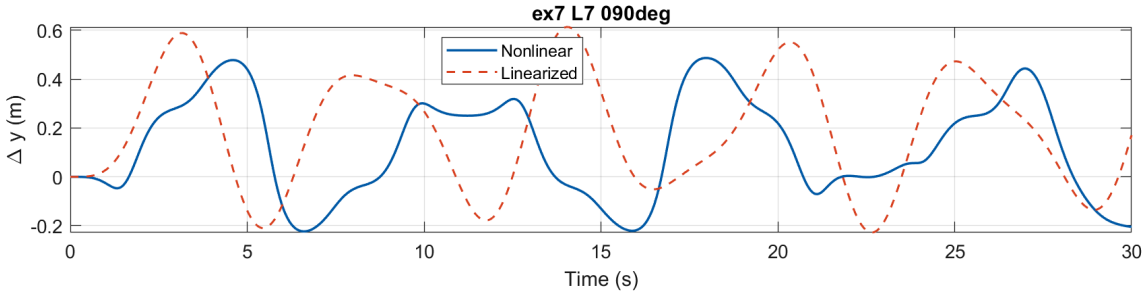


Figure 10: Δy: nonlinear and linearized time histories.

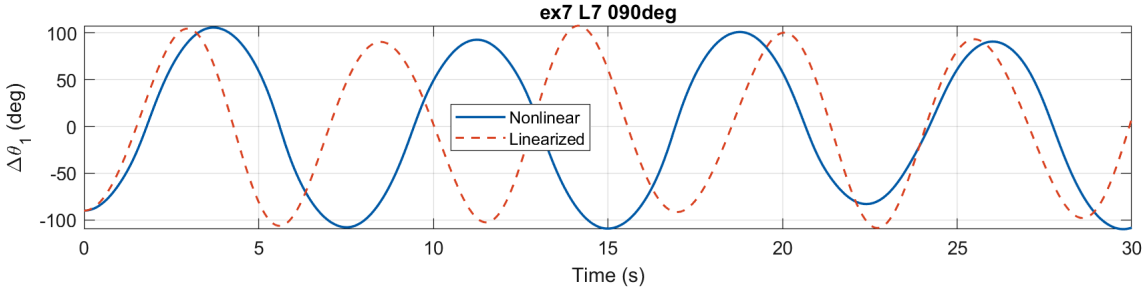


Figure 11: $\Delta\theta_1$: nonlinear and linearized time histories.

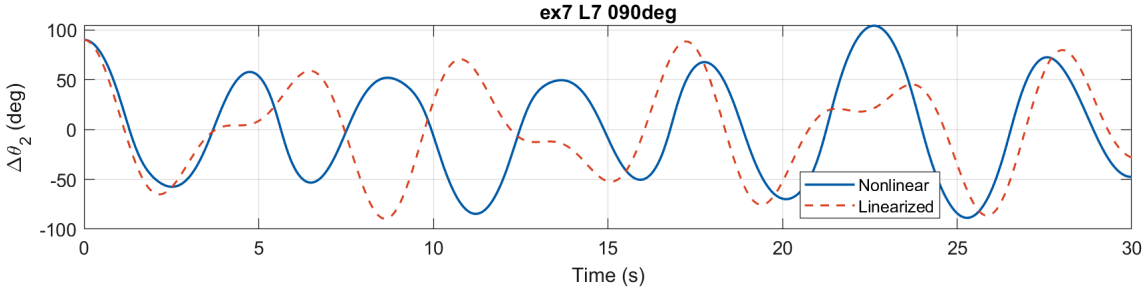


Figure 12: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code for L7, $\pm 90^\circ$: The following case-specific block uses the common local functions `simulate_case` and `export_case` listed in the source appendix.

```
p = [2 1 1 1 0.5 1]; % [m0 m1 m2 l1 l2 g]
model = cart_pendulum_model(p, 1);
alpha = 90; % degrees
z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];
options = odeset('RelTol',1e-10,'AbsTol',1e-12,'MaxStep',0.02);
r = simulate_case(model, z0, 30, 'ex7_L7_090deg', options);
export_case(r, outputDir, rootDir);
```

L7: $\pm 180^\circ$

Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0, -3.141593, 3.141593, 0, 0, 0]
```

$x^e = [0, 0, 0, 0, 0, 0]^T$, $x(0) = x^e + z(0)$.

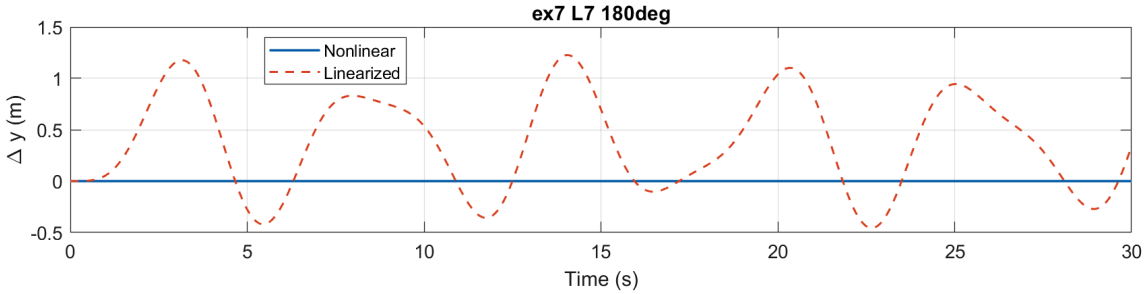


Figure 13: Δy: nonlinear and linearized time histories.

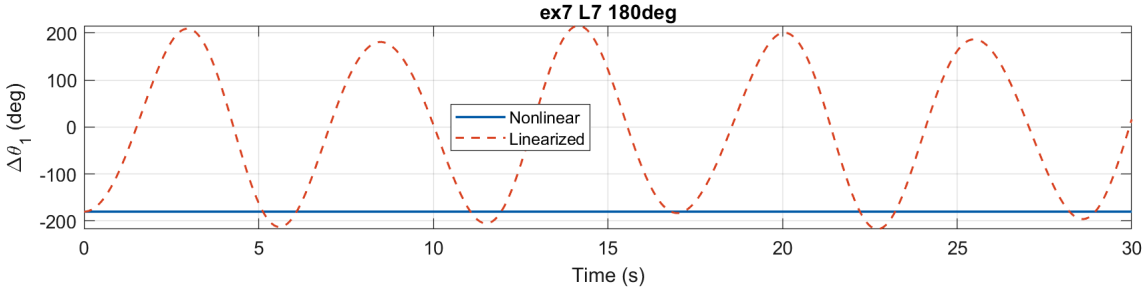


Figure 14: $\Delta\theta_1$: nonlinear and linearized time histories.

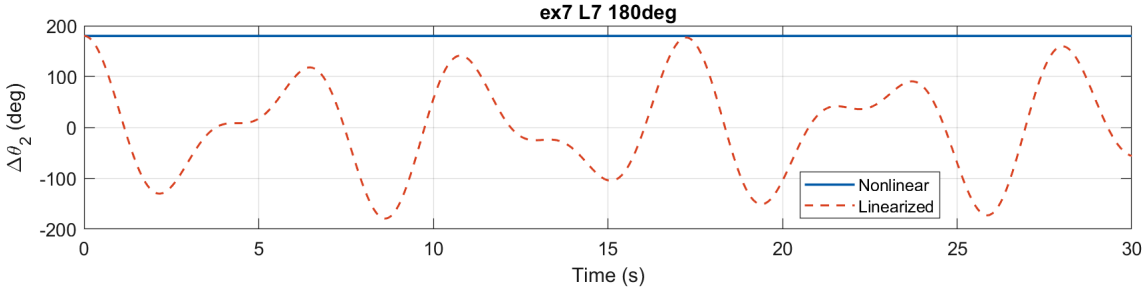


Figure 15: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code for L7, $\pm 180^\circ$: The following case-specific block uses the common local functions `simulate_case` and `export_case` listed in the source appendix.

```
p = [2 1 1 1 0.5 1]; % [m0 m1 m2 l1 l2 g]
model = cart_pendulum_model(p, 1);
alpha = 180; % degrees
z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];
options = odeset('RelTol',1e-10,'AbsTol',1e-12,'MaxStep',0.02);
r = simulate_case(model, z0, 30, 'ex7_L7_180deg', options);
export_case(r, outputDir, rootDir);
```

L8: $\pm 1^\circ$

Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0, -0.01745329, 0.01745329, 0, 0, 0]
```

$x^e = [0, 3.141593, 3.141593, 0, 0, 0]^T$, $x(0) = x^e + z(0)$.

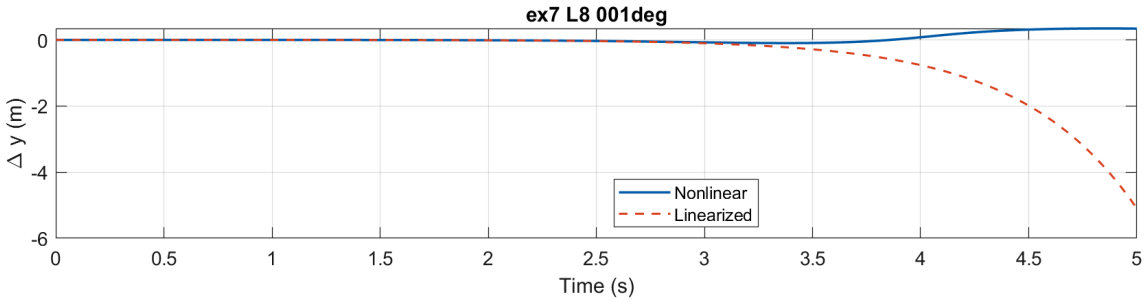


Figure 16: Δy: nonlinear and linearized time histories.

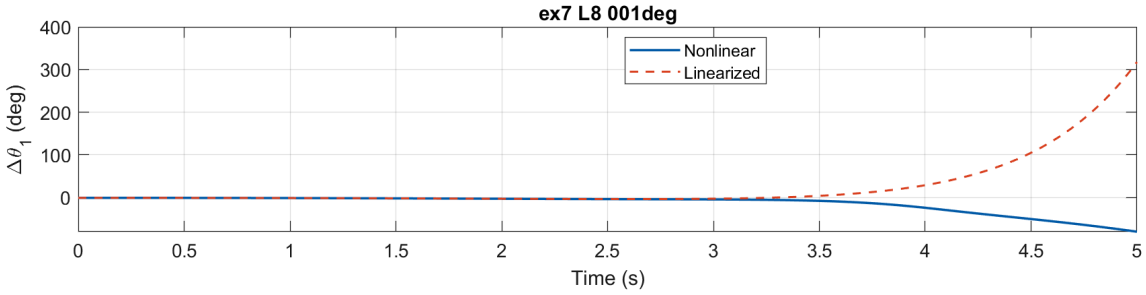


Figure 17: $\Delta\theta_1$: nonlinear and linearized time histories.

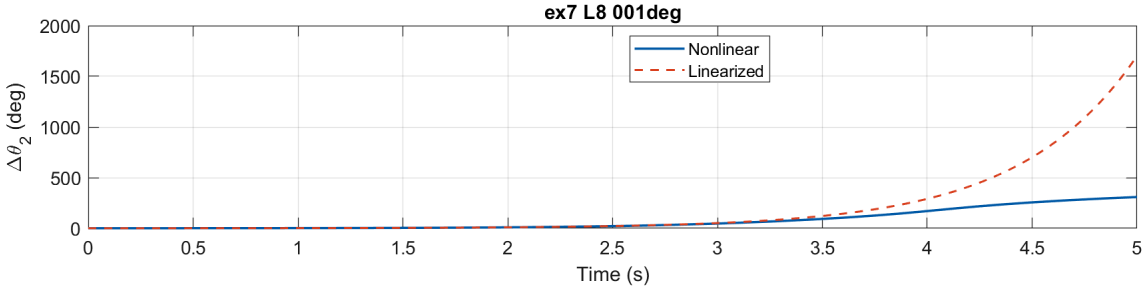


Figure 18: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code for L8, $\pm 1^\circ$: The following case-specific block uses the common local functions `simulate_case` and `export_case` listed in the source appendix.

```
p = [2 1 1 1 0.5 1]; % [m0 m1 m2 l1 l2 g]
model = cart_pendulum_model(p, 2);
alpha = 1; % degrees
z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];
options = odeset('RelTol',1e-10,'AbsTol',1e-12,'MaxStep',0.02);
r = simulate_case(model, z0, 5, 'ex7_L8_001deg', options);
export_case(r, outputDir, rootDir);
```

Exercise 8

This exercise refers to linearizations L7 and L8 of the two pendulum cart system.

(a) For L7 choose an initial state for the linearized system which results in a periodic solution for the linearized system.

Solution

Following the eigenvector method in the class notes, let

$$\omega = \sqrt{\frac{9-\sqrt{17}}{4}} \approx 1.1042, \qquad r = 2\omega^2 - 3, \qquad a = \frac{1+r}{2\omega^2}, \qquad c = \frac{\pi}{180}.$$

The factor c sets the maximum initial angular deviation to 1°. For L7, choose the eigenvalue $\lambda = j\omega$ and its eigenvector

$$v = \begin{bmatrix} a & 1 & r & j\omega a & j\omega & j\omega r \end{bmatrix}^T, \qquad A_{L7} v = j\omega v.$$

Choose $z(0) = c\,\mathrm{Re}(v) = c[a, 1, r, 0, 0, 0]^T$. Since A_{L7} is real,

$$z(t) = c\,\mathrm{Re}(e^{j\omega t} v) = c \begin{bmatrix} a\cos\omega t \\ \cos\omega t \\ r\cos\omega t \\ -a\omega\sin\omega t \\ -\omega\sin\omega t \\ -r\omega\sin\omega t \end{bmatrix}.$$

Hence, the solution is periodic with $T = 2\pi/\omega \approx 5.6903$ s. The nonlinear simulation uses $x(0) = x^e + z(0)$ with $x^e = 0$.

Comparison plots. Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[0.003138205, 0.01745329, -0.009800946, -1.197667e-19, 7.378286e-18, -2.67647e-18]
```

$x^e = [0, 0, 0, 0, 0, 0]^T$, $x(0) = x^e + z(0)$. The initial state is selected numerically from an eigenmode; see the source code. Linear states are deviations from equilibrium.

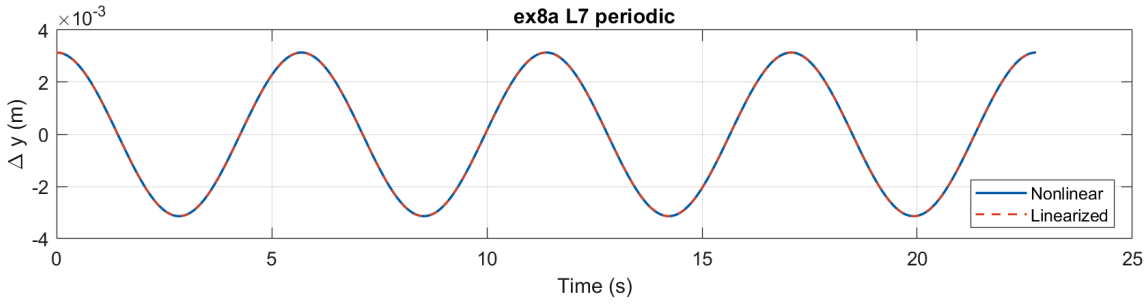


Figure 19: Δy: nonlinear and linearized time histories.

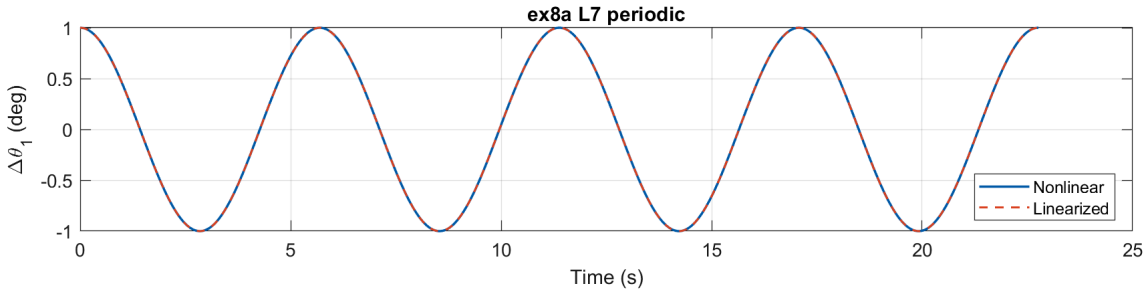


Figure 20: $\Delta\theta_1$: nonlinear and linearized time histories.

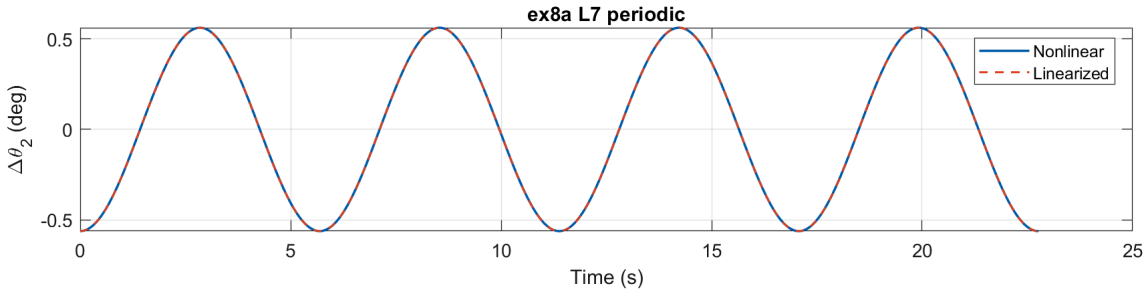


Figure 21: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code:

```
p = [2 1 1 1 0.5 1];
model = cart_pendulum_model(p, 1);
[V,D] = eig(model.A); lambda = diag(D);
idx = find(imag(lambda)>1e-8);
[~,j] = min(imag(lambda(idx))); j=idx(j);
v=V(:,j); [~,k]=max(abs(v(1:3)));
v=v*exp(-1i*angle(v(k)));
z0=scale_angles(real(v),1);
T=2*pi/imag(lambda(j));
r=simulate_case(model,z0,4*T,'ex8a_L7_periodic',options);
export_case(r,outputDir,rootDir);
```

(b) For L8 choose an initial state for the linearized system which results in a solution which asymptotically decays to zero for the linearized system.

Solution

For L8, choose the negative eigenvalue $\lambda_- = -\omega$ and

$$v_- = \begin{bmatrix} -a & 1 & r & \omega a & -\omega & -\omega r \end{bmatrix}^T, \qquad A_{L8} v_- = -\omega v_-.$$

Choose $z(0) = cv_-$. Then

$$z(t) = ce^{-\omega t} v_- \longrightarrow 0 \qquad (t \to \infty).$$

Thus the linearized state decays asymptotically to zero. For the nonlinear model, use $x(0) = x^e + cv_-$, where $x^e = [0, \pi, \pi, 0, 0, 0]^T$. The linear response below is evaluated as a single eigenmode to avoid numerical contamination by unstable modes. The nonlinear response need not remain on this linear stable eigendirection.

Comparison plots. Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[-0.003138205, 0.01745329, -0.009800946, 0.003465158, -0.01927166, 0.01082205]
```

$x^e = [0, 3.141593, 3.141593, 0, 0, 0]^T$, $x(0) = x^e + z(0)$. The initial state is selected numerically from an eigenmode; see the source code. Linear states are deviations from equilibrium.

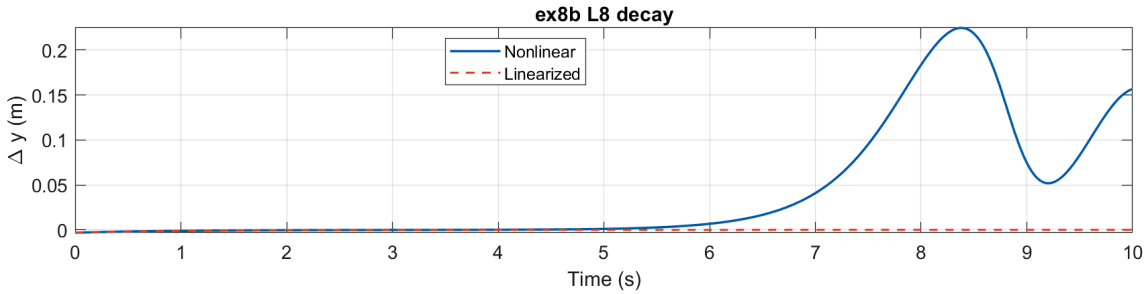


Figure 22: Δy: nonlinear and linearized time histories.

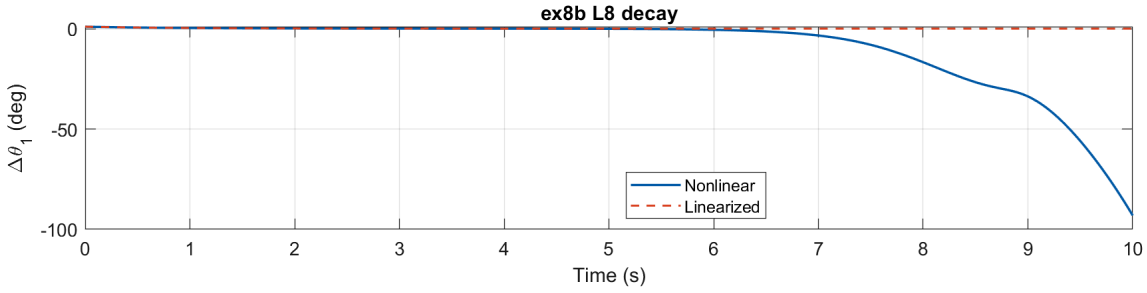


Figure 23: $\Delta\theta_1$: nonlinear and linearized time histories.

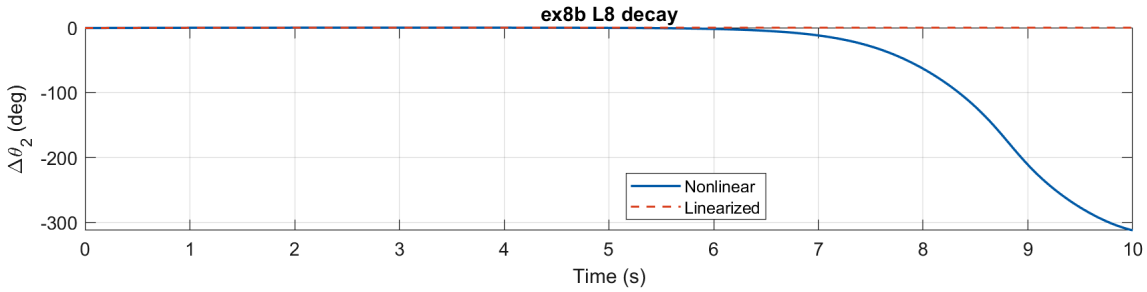


Figure 24: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code:

```
p = [2 1 1 1 0.5 1];
model = cart_pendulum_model(p, 2);
[V,D] = eig(model.A); lambda = diag(D);
idx=find(-1*real(lambda)>1e-8 & abs(imag(lambda))<1e-8);
[~,j]=min(abs(real(lambda(idx)))); j=idx(j);
v=real(V(:,j)); [~,k]=max(abs(v(2:3))); k=k+1;
if v(k)<0, v=-v; end
z0=scale_angles(v,1);
r=simulate_case(model,z0,10,'ex8b_L8_decay',options,real(lambda(j)));
export_case(r,outputDir,rootDir);
```

(c) For L8 choose an initial state for the linearized system which results in a solution whose magnitude grows exponentially for the linearized system.

Solution

For L8, choose the positive eigenvalue $\lambda_+ = \omega$ and

$$v_+ = \begin{bmatrix} -a & 1 & r & -\omega a & \omega & \omega r \end{bmatrix}^T, \qquad A_{L8} v_+ = \omega v_+.$$

Choose $z(0) = c v_+$. Then

$$z(t) = c e^{\omega t} v_+,$$

whose magnitude grows exponentially. The nonlinear model starts from $x(0) = x^e + c v_+$ with $x^e = [0, \pi, \pi, 0, 0, 0]^T$.

Comparison plots. Initial deviation $z(0)$ (radians for angles, SI units otherwise):

```
[-0.003138205, 0.01745329, -0.009800946, -0.003465158, 0.01927166, -0.01082205]
```

$x^e = [0, 3.141593, 3.141593, 0, 0, 0]^T$, $x(0) = x^e + z(0)$. The initial state is selected numerically from an eigenmode; see the source code. Linear states are deviations from equilibrium.

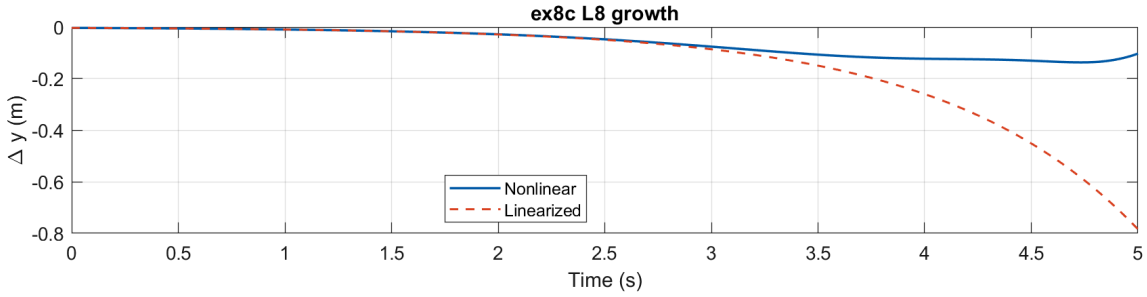


Figure 25: Δy: nonlinear and linearized time histories.

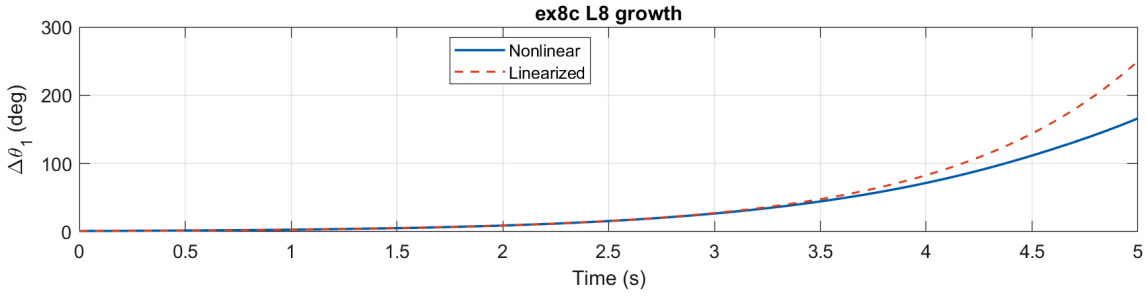


Figure 26: $\Delta\theta_1$: nonlinear and linearized time histories.

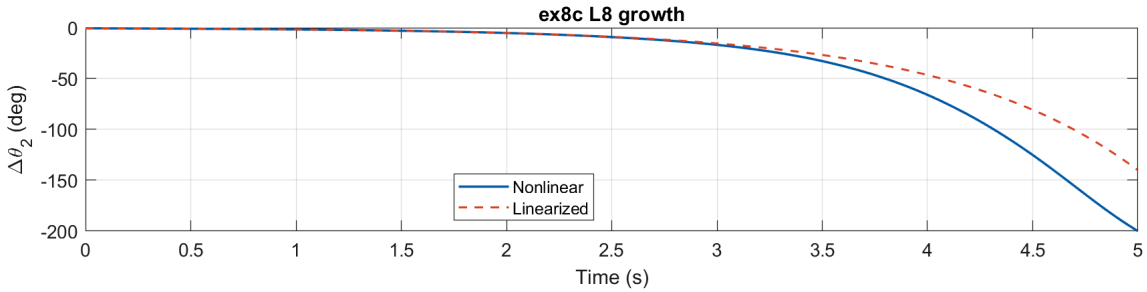


Figure 27: $\Delta\theta_2$: nonlinear and linearized time histories.

MATLAB code:

```
p = [2 1 1 1 0.5 1];
model = cart_pendulum_model(p, 2);
[V,D] = eig(model.A); lambda = diag(D);
idx=find(1*real(lambda)>1e-8 & abs(imag(lambda))<1e-8);
[~,j]=min(abs(real(lambda(idx)))); j=idx(j);
v=real(V(:,j)); [~,k]=max(abs(v(2:3))); k=k+1;
if v(k)<0, v=-v; end
z0=scale_angles(v,1);
r=simulate_case(model,z0,5,'ex8c_L8_growth',options,real(lambda(j)));
export_case(r,outputDir,rootDir);
```

In each case, simulate both the linearized system and the nonlinear system with initial conditions corresponding to your chosen initial state for the linearized system.

MATLAB Source Code

The following listings reproduce the source files used to generate the figures.

Main script

```
function hw4_ex7_ex8(exercise)
% Use hw4_ex7_ex8(7) to regenerate Exercise 7 only.
if nargin < 1, exercise = 0; end
assert(ismember(exercise, [0 7 8]), 'Use 0 (all), 7, or 8.');
clc;
% Homework 4, Exercises 7(b) and 8: nonlinear/linear comparisons.
% Run this script, with cart_pendulum_model.m in the same folder.
rootDir = fileparts(mfilename('fullpath'));
addpath(rootDir);
outputDir = fullfile(rootDir, 'figures');
if ~exist(outputDir, 'dir'), mkdir(outputDir); end

% Confirmed interpretation: use the Homework 1 parameter sets.
% Columns: m0, m1, m2, l1, l2, g.
P = [2 1 1   1 1    1;
     2 1 1   1 0.99 1;
     2 1 0.5 1 1    1;
     2 1 1   1 0.5  1];
models = cell(8, 1);
eigenvalues = zeros(6, 8);
for k = 1:8
    models{k} = cart_pendulum_model(P(ceil(k/2), :), 2-mod(k, 2));
    eigenvalues(:, k) = eig(models{k}.A);
    % Independent finite-difference check of the implemented Jacobian.
    J = zeros(6); h = 1e-6;
    for j = 1:6
        ej = zeros(6, 1); ej(j) = h;
        J(:, j) = (models{k}.rhs(0, models{k}.xe+ej) ...
                   -models{k}.rhs(0, models{k}.xe-ej))/(2*h);
    end
    assert(norm(J-models{k}.A, 'inf') < 1e-7);
end

options = odeset('RelTol', 1e-10, 'AbsTol', 1e-12, 'MaxStep', 0.02);
results = struct('id', {}, 'p', {}, 'A', {}, 'xe', {}, 'z0', {}, ...
    'x0', {}, 't', {}, 'zLinear', {}, 'xNonlinear', {}, ...
    'zNonlinear', {}, 'mode', {});

%% Exercise 7(b): opposite angular deviations from the equilibrium.
if exercise ~= 8
angles = [0 1 10 90 180];
for alpha = angles
    z0 = [0; deg2rad(-alpha); deg2rad(alpha); 0; 0; 0];
    id = sprintf('ex7_L7_%03ddeg', alpha);
    results(end+1) = simulate_case(models{7}, z0, 30, id, options);
end
z0 = [0; deg2rad(-1); deg2rad(1); 0; 0; 0];
results(end+1) = simulate_case(models{8}, z0, 5, ...
    'ex7_L8_001deg', options);

end

%% Exercise 8: select eigenmodes as in the class notes.
if exercise ~= 7
% 8(a): real part of an oscillatory mode, scaled to 1 degree peak angle.
[V7, D7] = eig(models{7}.A);
lambda7 = diag(D7);
candidates = find(imag(lambda7) > 1e-8);
[~, j] = min(imag(lambda7(candidates)));
j = candidates(j);
v = V7(:, j);
[~, pivot] = max(abs(v(1:3)));
v = v*exp(-1i*angle(v(pivot)));
```

```
z0 = scale_angles(real(v), 1);
period = 2*pi/imag(lambda7(j));
r = simulate_case(models{7}, z0, 4*period, 'ex8a_L7_periodic', options);
assert(norm(expm(models{7}.A*period)*z0-z0) < 1e-9);
r.mode = lambda7(j); results(end+1) = r;

% 8(b,c): negative/positive real modes of L8.
[V8, D8] = eig(models{8}.A);
lambda8 = diag(D8);
for signChoice = [-1 1]
    candidates = find(signChoice*real(lambda8) > 1e-8 ...
                      & abs(imag(lambda8)) < 1e-8);
    [~, j] = min(abs(real(lambda8(candidates))));
    j = candidates(j);
    v = real(V8(:, j));
    [~, pivot] = max(abs(v(2:3))); pivot = pivot+1;
    if v(pivot) < 0, v = -v; end
    z0 = scale_angles(v, 1);
    if signChoice < 0
        id = 'ex8b_L8_decay'; tf = 10;
    else
        id = 'ex8c_L8_growth'; tf = 5;
    end
    % Evaluate a pure real mode analytically to avoid numerical contamination
    % by the unstable modes when illustrating asymptotic linear decay.
    r = simulate_case(models{8}, z0, tf, id, options, real(lambda8(j)));
    r.mode = lambda8(j); results(end+1) = r;
end

end

%% Export separate plots, report fragments, and a reloadable workspace.
for k = 1:numel(results)
    export_case(results(k), outputDir, rootDir);
end
if exercise ~= 8
fid = fopen(fullfile(rootDir, 'simulation-ex7.tex'), 'w');
assert(fid ~= -1);
fprintf(fid, '\\subsection*{MATLAB time histories}\n');
fprintf(fid, ['Parameters P1--P4 follow Homework 1. ' ...
    'The two pendulum lengths are $l_1$ and $l_2$; for P4, ' ...
    '$l_1=1$, $l_2=0.5$.\n\n']);
fprintf(fid, ['State order: $x=[y,\\theta_1,\\theta_2,\\dot y,' ...
    '\\dot\\theta_1,\\dot\\theta_2]^T$. Both models start from the same ' ...
    'physical state $x(0)=x^e+z(0)$, with $u=0$. ' ...
    'Plots show deviations $z=x-x^e$; angles are in degrees and are not wrapped.\n\n']);
fprintf(fid, ['For Exercise 7(b), $z(0)=[0,-\\alpha,+\\alpha,0,0,0]^T$ ' ...
    '(angular entries converted from degrees to radians). ' ...
    'L7 uses $\\alpha=0,1,10,90,180$ deg and L8 uses $\\alpha=1$ deg. ' ...
    'L7 is simulated for 30 s and L8 for 5 s.\n\n']);
fprintf(fid, ['Nonlinear simulation: \\texttt{ode45}, relative tolerance ' ...
    '$10^{-10}$, absolute tolerance $10^{-12}$, maximum step 0.02 s. ' ...
    'Linear responses use the matrix exponential.\n']);
for k = 1:6
    fprintf(fid, '\\input{simulation-%s.tex}\n', results(k).id);
end
fclose(fid);
end
if exercise == 7
    workspaceFile = fullfile('build', 'hw4_ex7_workspace.mat');
    if ~exist(fullfile(rootDir, 'build'), 'dir'), mkdir(fullfile(rootDir, 'build')); end
elseif exercise == 8
    workspaceFile = fullfile('build', 'hw4_ex8_workspace.mat');
    if ~exist(fullfile(rootDir, 'build'), 'dir'), mkdir(fullfile(rootDir, 'build')); end
else
    workspaceFile = 'hw4_workspace.mat';
end
save(fullfile(rootDir, workspaceFile), ...
    'P', 'models', 'eigenvalues', 'results', 'options');
fprintf('Finished: %d cases, %d separate plots.\n', numel(results), 3*numel(results));
fprintf('Workspace saved to %s\n', fullfile(rootDir, workspaceFile));
end
```

```
function z = scale_angles(v, degrees)
z = v*(deg2rad(degrees)/max(abs(v(2:3))));
end

function r = simulate_case(model, z0, tf, id, options, mode)
t = linspace(0, tf, 1501).';
[t, xNonlinear] = ode45(model.rhs, t, model.xe+z0, options);
zLinear = zeros(numel(t), 6);
if nargin == 6
    assert(norm(model.A*z0-mode*z0) < 1e-9);
    zLinear = exp(t*mode)*z0.';
else
    for k = 1:numel(t)
        zLinear(k, :) = (expm(model.A*t(k))*z0).';
    end
end
r = struct('id', id, 'p', model.p, 'A', model.A, 'xe', model.xe, ...
    'z0', z0, 'x0', model.xe+z0, 't', t, 'zLinear', zLinear, ...
    'xNonlinear', xNonlinear, 'zNonlinear', xNonlinear-model.xe.', ...
    'mode', NaN);
assert(all(isfinite(xNonlinear(:))) && all(isfinite(zLinear(:))));
assert(norm(r.zNonlinear(1, :).'-z0) < 1e-12);
end

function export_case(r, outputDir, rootDir)
names = {'y', 'theta1', 'theta2'};
labels = {'\Delta y (m)', '\Delta\theta_1 (deg)', '\Delta\theta_2 (deg)'};
captionNames = {'$\Delta y$', '$\Delta\theta_1$', '$\Delta\theta_2$'};
fid = fopen(fullfile(rootDir, ['simulation-' r.id '.tex']), 'w');
assert(fid ~= -1);
if startsWith(r.id, 'ex7')
    tokens = regexp(r.id, 'ex7_L(\d+)_(\d+)deg', 'tokens', 'once');
    caseNumber = str2double(tokens{1});
    alpha = str2double(tokens{2});
    if alpha == 0
        angleTitle = '0';
    else
        angleTitle = sprintf('\\pm%d', alpha);
    end
    fprintf(fid, '\\clearpage\n\\subsubsection*{L%d: $%s^\\circ$}\n', caseNumber, angleTitle);
else
    fprintf(fid, '\\par\\medskip\n\\textbf{Comparison plots.}\n');
end
fprintf(fid, 'Initial deviation $z(0)$ (radians for angles, SI units otherwise):\n');
fprintf(fid, '\\begin{center}\\scriptsize\\texttt{[');
fprintf(fid, '%.7g, ', r.z0(1:5)); fprintf(fid, '%.7g]}\\end{center}\n', r.z0(6));
fprintf(fid, '$x^e=[0,%.7g,%.7g,0,0,0]^T$, $x(0)=x^e+z(0)$.\n', r.xe(2:3));
if startsWith(r.id, 'ex8')
    fprintf(fid, ['The initial state is selected numerically from an eigenmode; ' ...
        'see the source code. Linear states are deviations from equilibrium.\n']);
end
for j = 1:3
    scale = 1; if j > 1, scale = 180/pi; end
    fig = figure('Visible', 'off', 'Color', 'w', ...
        'Position', [100 100 1000 270]);
    ax = axes(fig, 'Position', [0.12 0.24 0.84 0.60]);
    plot(ax, r.t, scale*r.zNonlinear(:, j), '-', ...
        'Color', [0 0.35 0.65], 'LineWidth', 1.4); hold(ax, 'on');
    plot(ax, r.t, scale*r.zLinear(:, j), '--', ...
        'Color', [0.85 0.25 0.12], 'LineWidth', 1.2);
    grid(ax, 'on'); xlabel(ax, 'Time (s)'); ylabel(ax, labels{j});
    title(ax, strrep(r.id, '_', ' '));
    legend(ax, 'Nonlinear', 'Linearized', 'Location', 'best');
    set(ax, 'FontSize', 11, 'Color', 'w', 'XColor', [0.1 0.1 0.1], ...
        'YColor', [0.1 0.1 0.1]);
    exportgraphics(fig, fullfile(outputDir, [r.id '_' names{j} '.png']), ...
        'Resolution', 180, 'BackgroundColor', 'white');
    savefig(fig, fullfile(outputDir, [r.id '_' names{j} '.fig']));
    close(fig);
    fprintf(fid, '\\begin{figure}[!ht]\n\\centering\n');
    fprintf(fid, '\\includegraphics[width=0.94\\linewidth]{figures/%s_%s.png}\n', r.id, names{j});
    fprintf(fid, '\\caption{%s: nonlinear and linearized time histories.}\n\\end{figure}\n', captionNames{j
    });
```

```
end
if startsWith(r.id, 'ex7')
    fprintf(fid, '\n\\textbf{MATLAB code for L%d, $%s^\\circ$:}\n', caseNumber, angleTitle);
    fprintf(fid, ['The following case-specific block uses the common local functions ' ...
        '\\texttt{simulate\\_case} and \\texttt{export\\_case} listed in the source appendix.\n']);
    fprintf(fid, '\\begin{lstlisting}\n');
    fprintf(fid, 'p = [2 1 1 1 0.5 1]; %% [m0 m1 m2 l1 l2 g]\n');
    fprintf(fid, 'model = cart_pendulum_model(p, %d);\n', 1 + (caseNumber == 8));
    fprintf(fid, 'alpha = %d; %% degrees\n', alpha);
    fprintf(fid, 'z0 = [0; -deg2rad(alpha); deg2rad(alpha); 0; 0; 0];\n');
    fprintf(fid, 'options = odeset(''RelTol'',1e-10,''AbsTol'',1e-12,''MaxStep'',0.02);\n');
    fprintf(fid, 'r = simulate_case(model, z0, %g, ''%s'', options);\n', r.t(end), r.id);
    fprintf(fid, 'export_case(r, outputDir, rootDir);\n');
    fprintf(fid, '\\end{lstlisting}\n');
end
if startsWith(r.id, 'ex8')
    fprintf(fid, '\\Needspace{5cm}\n\\textbf{MATLAB code:}\n\\begin{lstlisting}\n');
    fprintf(fid, 'p = [2 1 1 1 0.5 1];\n');
    isPeriodic = startsWith(r.id, 'ex8a');
    fprintf(fid, 'model = cart_pendulum_model(p, %d);\n', 2-isPeriodic);
    fprintf(fid, '[V,D] = eig(model.A); lambda = diag(D);\n');
    if isPeriodic
        fprintf(fid, 'idx = find(imag(lambda)>1e-8);\n[~,j] = min(imag(lambda(idx))); j=idx(j);\n');
        fprintf(fid, 'v=V(:,j); [~,k]=max(abs(v(1:3)));\nv=v*exp(-1i*angle(v(k)));\nz0=scale_angles(real(v)
     ,1);\n');
        fprintf(fid, 'T=2*pi/imag(lambda(j));\nr=simulate_case(model,z0,4*T,''%s'',options);\n', r.id);
    else
        signChoice = 1; if startsWith(r.id, 'ex8b'), signChoice = -1; end
        fprintf(fid, 'idx=find(%d*real(lambda)>1e-8 & abs(imag(lambda))<1e-8);\n', signChoice);
        fprintf(fid, '[~,j]=min(abs(real(lambda(idx)))); j=idx(j);\n');
        fprintf(fid, 'v=real(V(:,j)); [~,k]=max(abs(v(2:3))); k=k+1;\nif v(k)<0, v=-v; end\nz0=scale_angles
     (v,1);\n');
        fprintf(fid, 'r=simulate_case(model,z0,%g,''%s'',options,real(lambda(j)));\n', r.t(end), r.id);
    end
    fprintf(fid, 'export_case(r,outputDir,rootDir);\n\\end{lstlisting}\n');
end
fclose(fid);
end
```

Model function

```
function model = cart_pendulum_model(p, equilibrium)
% Parameters: [m0 m1 m2 l1 l2 g]. State: [y th1 th2 yd w1 w2].
% The linear state is the deviation z = x - xe; input u = 0.
m0 = p(1); m1 = p(2); m2 = p(3);
l1 = p(4); l2 = p(5); g = p(6);
if equilibrium == 1
    xe = zeros(6, 1);
    c = 1;
else
    xe = [0; pi; pi; 0; 0; 0];
    c = -1;
end
M = [m0+m1+m2, -m1*l1*c, -m2*l2*c;
     -m1*l1*c, m1*l1^2, 0;
     -m2*l2*c, 0, m2*l2^2];
K = diag([0, m1*l1*g*c, m2*l2*g*c]);
model.A = [zeros(3), eye(3); -(M\K), zeros(3)];
model.B = [zeros(3, 1); M\[1; 0; 0]];
model.xe = xe;
model.p = p;
model.rhs = @(t, x) nonlinear_rhs(t, x, p);
end

function dx = nonlinear_rhs(~, x, p)
m0 = p(1); m1 = p(2); m2 = p(3);
l1 = p(4); l2 = p(5); g = p(6);
s1 = sin(x(2)); s2 = sin(x(3));
c1 = cos(x(2)); c2 = cos(x(3));
% Preserve exact equilibria at integer multiples of pi in floating point.
if abs(s1) < 8*eps, s1 = 0; end
if abs(s2) < 8*eps, s2 = 0; end
ydd = (-m1*g*c1*s1 - m2*g*c2*s2 ...
       -m1*l1*s1*x(5)^2 - m2*l2*s2*x(6)^2) ...
      / (m0 + m1*s1^2 + m2*s2^2);
dx = [x(4:6); ydd; (c1*ydd-g*s1)/l1; (c2*ydd-g*s2)/l2];
end
```